MM

Raw Paper

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1 Fourier Transform

$$f(t) = \frac{1}{\tau\sqrt{2\pi}} exp\left(-\frac{t^2}{2\tau^2}\right) \tag{1}$$

$$\tilde{f}(\omega) = \frac{1}{\tau 2\pi}\int_{-\infty}^{\infty} exp\left(-\frac{t^2}{2\tau^2}\right) exp(-\mathrm{i}\omega t)\, dt \qquad = \tag{2}$$

2 Toturial

$$\begin{gathered}\langle f, g\rangle \\ T^\dagger \\ L^2(\mathbb{R}) \\ \langle f, g\rangle = \int_{\mathbb{R}} f(x)^* g(x)\, dx \\ T = \frac{d^2}{dx^2} + 2c\frac{d}{dx} + 1, \qquad Ty = f.\end{gathered} \tag{3}$$

$$\begin{gathered}\sum_{k=0}^{\infty} \\ \sum_{k=0}^{\infty} a_k x^k \\ a_{k+2} = \cdots \quad \text{for } k = 0, 1, 2, \ldots \\ \delta_{nm}\end{gathered} \tag{4}$$

$$y(x) = \sum_{k=0}^{\infty} a_k x^k, \qquad a_{k+2} = \frac{k(k+1) - n(n+1)}{(k+1)(k+2)} a_k.$$

3 Equations...

$$W(y_1, y_2) = y_1 y_2' - y_1' y_2 \tag{5}$$

$$G(x,t) = \frac{y_1(t)y_2(x) - y_1(x)y_2(t)}{W(t)} \tag{6}$$

$$y(x) = \int_{x_0}^{x} G(x,t) f(t)\, dt \tag{7}$$

4 Hermitian, unitary and commutator notation

$$\begin{gathered} T \circ s \\ [T, S] \\ T^\dagger = T \\ U^\dagger U = I \\ Tv = \lambda v \\ \Longleftarrow \\ \mathbb{R} \\ [T, S] := T \circ S - S \circ T, \qquad T^\dagger = T. \\ U^\dagger U = U U^\dagger = I \end{gathered} \tag{8}$$

5 Matrices and vectors

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \qquad v = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$$

$$A^{\mathsf{T}}$$

$$\det(A)$$

$$H_n(x) P_n(x) L_n(x) \mathcal{L} J_\nu(x) \Gamma(x) \frac{d^n}{dx^n} w(x) \langle H_n, H_m \rangle = 0$$

$$H_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n} e^{-x^2}$$

$$y'' - 2xy' + 2ny = 0 \tag{9}$$

$$\langle H_n, H_m \rangle = \int_{-\infty}^{\infty} e^{-x^2} H_n(x) H_m(x)\, dx = \sqrt{\pi}\, 2^n n!\, \delta_{nm}$$

$$x^2 y'' + xy' + (x^2 - \nu^2) y = 0, \qquad J_\nu(x) = \left(\frac{x}{2}\right)^\nu \sum_{k=0}^{\infty} \frac{(-1)^k}{k!\, \Gamma(k + \nu + 1)} \left(\frac{x}{2}\right)^{2k}$$

$$T_{\mathrm{SL}} y = \frac{1}{w(x)} \left[\frac{d}{dx} \left(p(x) \frac{dy}{dx} \right) + q(x) y \right]$$

$$T_{\mathrm{SL}} y = \lambda y.$$

6 Harmonci oscillator notation

$\epsilon, \hbar, \Psi, \psi, a^\dagger, a, N$

$$H = \frac{1}{2} \left(-\frac{d^2}{dx^2} + x^2 \right), \qquad H|n\rangle = \left(n + \frac{1}{2} \right) |n\rangle$$

$$\text{for all } x$$

$$a = \frac{1}{\sqrt{2}} \left(x + \frac{d}{dx} \right), \qquad a^\dagger = \frac{1}{\sqrt{2}} \left(x - \frac{d}{dx} \right) \tag{10}$$

$$\{ x \in \mathbb{R} \mid x > 0 \}$$

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7 Orthogonal Polynomials

$$\begin{aligned} \phi_0 &= 1, \\ \langle \phi_0, \phi_0 \rangle^{\frac{1}{2}} &= \left(\int_{-1}^{1} \mathrm{d}u \right) = \sqrt{2}, \\ \hat{\phi}_0 &= \frac{1}{\sqrt{2}}, \end{aligned} \tag{11}$$

The second polynomial is given by

$$\begin{aligned} \phi_1 &= y_1 - \hat{\phi}_0 \langle \hat{\phi}_0 | y_1 \rangle \\ \text{it is easily to see that } \langle \hat{\phi}_0 | y_1 \rangle &= 0, \\ \text{so } \phi_1 &= x, \\ \hat{\phi}_1 = \phi_1 \left(\int_{-1}^{1} u \times u \, \mathrm{d}u \right)^{-\frac{1}{2}} &= \sqrt{\frac{3}{2}}\, x. \end{aligned} \tag{12}$$

The third polynomial is likely given by

$$\begin{aligned} \phi_2 &= y_2 - \hat{\phi}_1 \langle \hat{\phi}_1 | y_2 \rangle - \hat{\phi}_0 \langle \hat{\phi}_0 | y_2 \rangle, \\ &= x^2 - 0 - \sqrt{\frac{1}{2}} \left(\int_{-1}^{1} \sqrt{\frac{1}{2}} \times x^2 \, \mathrm{d}u \right) = x^2 - \frac{1}{3}, \end{aligned} \tag{13}$$

which, on normalising gives

$$\hat{\phi}_2 = \phi_2 \left(\int_{-1}^{1} (x^2 - \frac{1}{3})^2 \, \mathrm{d}u \right)^{-\frac{1}{2}} = \sqrt{\frac{5}{8}} \left(3x^2 - 1 \right). \tag{14}$$

8 Legendre Polynomials

In the Hilbert space $L^2([-1,1])$, Legendre polynomials' Rodriguez formula reads

$$P_n(x) = \frac{1}{2^n n!}\frac{\mathrm{d}^n}{\mathrm{d}x^n}(x^2-1)^n \quad \Longrightarrow \quad K_n = 2^n n!, \tag{15}$$

thus, the Legendre differential equation is given by

$$(1-x^2)y'' - 2xy' + n(n+1)y = 0. \tag{16}$$

We also have the generating function of Legendre polynomials defined as

$$G(x,z) = \sum_{n=0}^{\infty} L_n(x) z^n \tag{17}$$

9 Hermite Polynomials

TheHermite polynomials are defined on the interval $[a,b] = [-\infty,\infty] = \mathbb{R}$, they are denoted by H_n with Rodriguez fromula reads

$$H_n(x) = (-1)^n \mathrm{e}^{x^2}\frac{\mathrm{d}}{\mathrm{d}x^n}\,\mathrm{e}^{-x^2}, \tag{18}$$

from this formula, it is easy to read off the coefficients k_n and k'_n of the leading and subleading monomials x^n and x^{n-1} in H_n as

$$k_n = 2^n, \qquad k'_n = 0. \tag{19}$$

The differential equation for Hemite polynomials is

$$y'' - 2xy' + 2ny = 0. \tag{20}$$

The generation function

$$G(x,z) = \sum_{n=0}^{\infty} H_n(x)\frac{z^n}{n!} = \mathrm{e}^{2xz-z^2}. \tag{21}$$

10 Gamma Function

The Gamma function is defined by

$$\Gamma(x) = \int_0^{\infty} e^{-t}t^{x-1}\,\mathrm{d}t, \tag{22}$$

and satisfies the functional equation

$$\Gamma(x+1) = x\Gamma(x). \tag{23}$$

And the Gamma function is related to the factorial function by

$$\Gamma(n+1) = n!. \tag{24}$$

11 Bessel Function

The Bessel function of the first kind is defined by

$$J_{\pm\nu}(x) := \left(\frac{x}{2}\right)^{\pm\nu} \sum_{k=0}^{\infty} \frac{(-1)^k}{k!\,\Gamma(k \pm \nu + 1)} \left(\frac{x}{2}\right)^{2k}, \tag{25}$$

The Bessel function of the second kind is defined by

$$N_\nu(x) = \frac{J_\nu(x)\cos(\nu\pi) - J_{-\nu}(x)}{\sin(\nu\pi)}. \tag{26}$$